

Computer Networks – FULL EXAM NOTES

These notes are prepared strictly for university examinations by referring to all provided lecture PDFs. They include definitions, explanations, advantages, disadvantages, working principles, and guidance on diagrams to draw in exams.

1. Switching and Router Architecture

1.1 What is Switching?

Switching is the process of transferring data packets or frames from an incoming port to the appropriate outgoing port. Switching ensures that data reaches the correct destination efficiently within a network device such as a switch or router.

1.2 Basic Router Architecture

A router consists of three major components: input ports, switching fabric, and output ports. Routing decisions are handled by the control plane, while packet forwarding is handled by the data plane.

- Control Plane: Runs routing protocols (OSPF, RIP, BGP). Operates in milliseconds.
- Data Plane: Performs fast packet forwarding using hardware. Operates in nanoseconds.
- Separation improves performance and scalability.

1.3 Input Port Functions

- Physical layer processing (bit-level reception)
- Data link layer processing (frame decapsulation)
- Lookup and forwarding decision using forwarding table
- Queueing when packets arrive faster than switching rate

■ Exam Diagram: Generic Router Architecture with Input Ports, Switching Fabric, Output Ports, and Routing Processor.

2. Switching Fabrics

2.1 Switching via Memory

In memory-based switching, packets are copied from the input port to system memory and then forwarded to the output port. This method is simple but limited by memory bandwidth.

- Used in first-generation routers
- Two memory accesses per packet (input and output)
- Performance bottleneck at high speeds

2.2 Switching via Bus

In bus-based switching, all input and output ports share a common bus. Only one packet can be transmitted at a time, leading to bus contention.

2.3 Switching via Interconnection Network

Interconnection networks such as crossbar and Banyan networks allow multiple packets to be switched in parallel. This provides high throughput and scalability.

■ Exam Diagram: Memory, Bus, and Crossbar Switching Fabric comparison.

3. Virtual Local Area Network (VLAN)

3.1 What is VLAN?

A VLAN is a logical grouping of network devices that creates separate broadcast domains within a switch. Devices in the same VLAN behave as if they are on the same physical network.

3.2 Why VLAN?

- Reduces broadcast traffic
- Improves network performance
- Enhances security
- Simplifies network management

3.3 VLAN Membership

- Static VLAN: VLAN manually assigned to switch ports
- Dynamic VLAN: VLAN assigned automatically using VMPS

■ Exam Diagram: VLAN segmentation showing multiple VLANs on a single switch.

4. Spanning Tree Protocol (STP)

4.1 Need for STP

Redundant links in Layer 2 networks can cause loops, broadcast storms, and duplicate frames. STP prevents these problems by creating a loop-free topology.

4.2 Root Bridge Selection

Each switch has a Bridge ID (Bridge Priority + MAC Address). The switch with the lowest Bridge ID becomes the root bridge.

4.3 STP Port Roles

- Root Port: Lowest cost path to root bridge
- Designated Port: Forwarding port on a segment
- Blocked Port: Prevents loops

■ Exam Diagram: STP topology showing root bridge and blocked links.

5. Internet Protocol Version 6 (IPv6)

5.1 Need for IPv6

IPv4 address exhaustion and modern networking requirements led to the development of IPv6.

5.2 Features of IPv6

- 128-bit address space
- No broadcast (uses multicast)
- Simplified header
- Built-in security (IPSec support)

5.3 IPv6 Header

IPv6 has a fixed 40-byte header with fields such as Version, Traffic Class, Flow Label, Payload Length, Next Header, and Hop Limit.

■ Exam Diagram: IPv6 header format.

6. Router Configuration and ICMP

6.1 Router Configuration Files

- Startup Configuration: Stored in NVRAM
- Running Configuration: Stored in RAM

6.2 ICMP

Internet Control Message Protocol is used for error reporting and diagnostics such as ping and traceroute.

7. Optical Transmission and Multiplexing

7.1 Optical Transmission

Data is transmitted as light pulses through optical fiber. Fiber optics provide high bandwidth and immunity to electromagnetic interference.

7.2 Types of Fiber

- Multimode Step Index
- Multimode Graded Index
- Single Mode Fiber

8. Network Delay and Packet Loss

- Processing Delay
- Queuing Delay
- Transmission Delay
- Propagation Delay

■ Exam Diagram: Four types of delay with router and links.

9. WAN Technologies

- Point-to-Point Links
- Circuit Switching (ISDN)
- Packet Switching (Frame Relay, ATM)
- Virtual Circuits (SVC and PVC)

10. Data Link Control and ARQ Protocols

10.1 Flow Control

Flow control prevents a fast sender from overwhelming a slow receiver.

10.2 Error Control

Error control ensures reliable delivery using retransmission mechanisms known as ARQ.

10.3 ARQ Protocols

- Stop-and-Wait ARQ
- Go-Back-N ARQ
- Selective Repeat ARQ

■ Exam Diagram: Stop-and-Wait and Sliding Window operation.

— END OF FULL EXAM NOTES —